www.tntextbooks.in

maintaining electrolytes, osmotic pressure and blood pressure. Adrenal androgen plays a role in hair growth in the axial region, pubis and face during puberty.

The **adrenal medulla** secretes the hormones adrenalin and noradrenalin and are referred as "3F hormone" (fight, flight and fright hormone). Adrenalin increases liver glycogen breakdown into glucose and increases the release of fatty acids from fat cells. During emergency it increases heart beat rate and blood pressure. It stimulates the smooth muscles of cutaneous and visceral arteries to decrease blood flow. It increases blood flow to the skeletal muscles thereby increases the metabolic rate of skeletal muscles, cardiac muscles and nervous tissue.

11.2.8 Pancreas

Pancreas is a composite gland which performs both exocrine and endocrine functions. It is located just below the stomach as a leaf like structure. The pancreas is composed of two major tissues such the acini and islets of langerhans. Acini secretes digestive enzymes and the islets of langerhans secretes hormones like insulin and glucagon. Human pancreas has one to two million islets of langerhans. In each islet about 60% cells are beta cells, 25% cells are alpha cells and 10% cells are delta cells. The alpha cells secrete glucagon, the beta cells secrete insulin and delta cells secrete somatostatin.

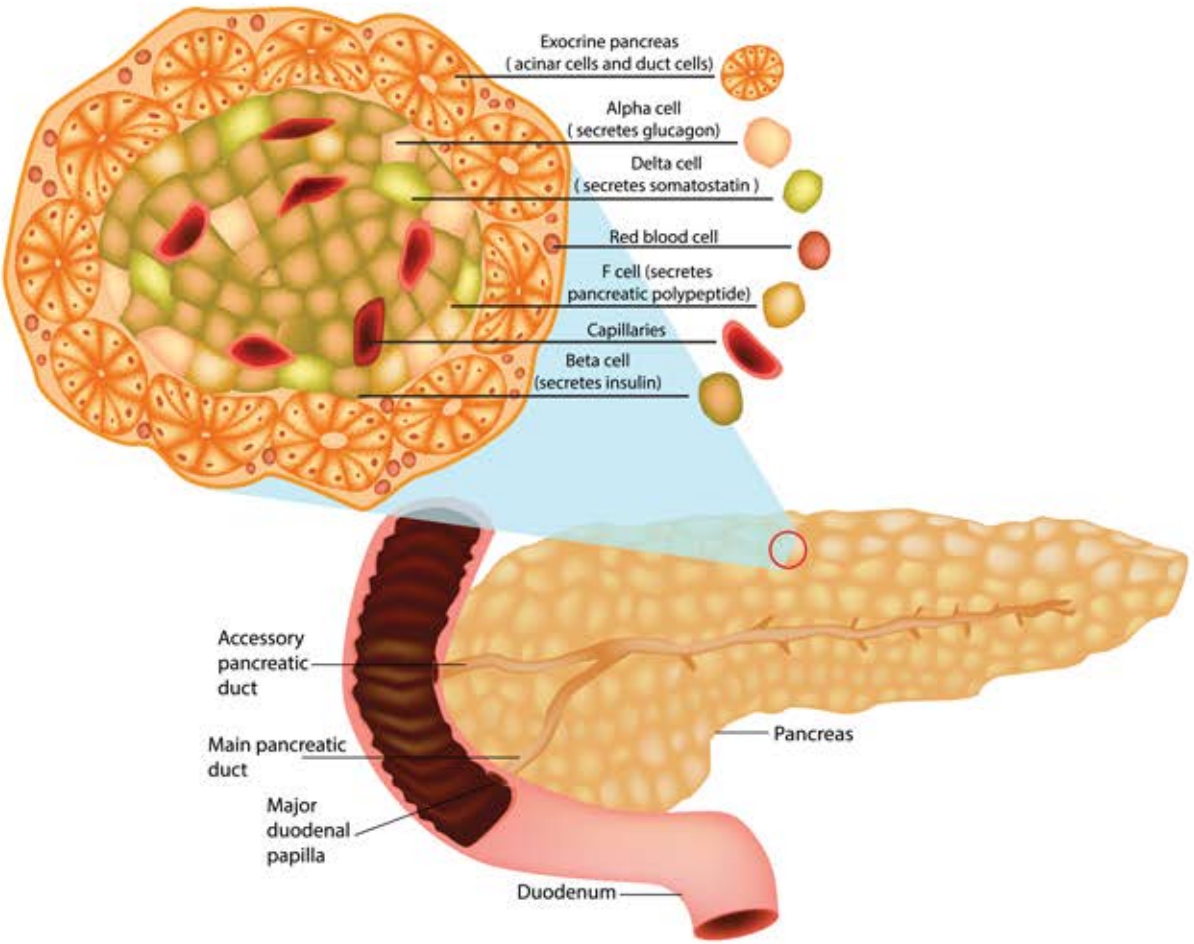


Figure 11.7: Structure of Islets of langerhans (pancreas)